



(ACT,AD,AM,CB,CC,CS,EC,EE,IS,RI,VLSI)

Max. Marks: 100

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

20 Marks

- 1-

10. What is the output of the following C program?

```
#include<stdio.h>
int main()
{
    int x[5] = {10,20,30};
    printf("%d",x[3]);
    return 0;
}
```

- A) 30
B) error
C) 0
D) Garbage value
11. Which of the following is a correct format for declaration of function?
A) return-type function-name(argument type);
B) return-type function-name(argument type){}
C) return-type (argument type)function-name;
D) (argument type)function-name;
12. Which of the following is not a basic element of a user defined function
A) Function declaration
B) Function call
C) Function definition
D) Return type
13. Which of the following is the collection of different data types?
A) Structure
B) String
C) Array
D) pointer
14. If the two strings are identical, then strcmp() function returns
A) 1
B) 0
C) -1
D) true
15. Which of the following function is used to find the first occurrence of a character in a string.
A) strchr()
B) strrchr()
C) strstr()
D) strcat()
16. What is the size of a C structure?
A) C structure is always 128 bytes
B) Size of C structure is the size of largest elements
C) Size of C structure is the total bytes of all elements of structure
D) Size of C structure is the size of smallest elements
17. How do you declare a pointer variable that can store the address of a character variable in C?
A) int *ptr
B) char ptr;
C) ptr *char
D) char *ptr;
18. What is the result of the expression ptr++ in C, where ptr is a pointer?
A) Increase the value of ptr by 1
B) It is not a valid operation
C) Decrease the value of ptr by 1
D) Move ptr to the next memory location
19. Which of the following true about FILE *fp
A) FILE is a keyword in C for representing files
B) FILE is a stream
C) FILE is a buffered stream
D) FILE is a structure and fp is a pointer to the structure of FILE type
20. Which of the following function is used to create a file in C?
A) fopen()
B) fseek()
C) fseek()
D) fclose()

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Discuss implicit and explicit type conversions with example.	06	L*2	1	1
b)	Differentiate while and do-while loop. Write a C program to find the factorial of a given number using for loop.	06	L2	2	2
c)	Explain the following constructs in C.				
i)	break				
ii)	continue	04	L2	2	1

- | | | | | | |
|-------|---|----|----|---|---|
| 2. a) | Evaluate the following expressions: | | | | |
| i) | $(a/b-c) \&\& (d\%e) \parallel (f/a) != d < !f$, where $a=10, b=2, c=4, d=7, e=3$ and $f=0$. | 06 | L2 | 1 | 2 |
| ii) | $b = a++ + ++a$, where $a=5$. | | | | |
| b) | What is an identifier in C? Write rules for writing a valid identifier. Give Examples for valid identifiers. | 05 | L2 | 1 | 1 |
| c) | Explain conditional operator with syntax. Write a program to find the greatest of three numbers using conditional operator. | 05 | L3 | 2 | 2 |
| 3. a) | Discuss logical and bitwise operators in C with program to illustrate each. | 08 | L2 | 1 | 2 |
| b) | List out the basic datatypes in C with its keyword and number of bytes used. | 04 | L1 | 1 | 1 |
| c) | Build a C program to find the GCD of two numbers using Euclid's method. | 04 | L3 | 2 | 2 |

Unit – II

- | | | | | | |
|-------|---|----|----|---|---|
| 4. a) | Build a C program to read a matrix of order $m \times n$ and find its transpose. | 06 | L3 | 3 | 2 |
| b) | What is string? Write a C program to read 2 strings and concatenate them. Find the length of concatenated string and display it. | 06 | L3 | 4 | 1 |
| c) | Define an array. Explain declaration and initialization of 1-D array. | 04 | L2 | 3 | 1 |
| 5. a) | Explain any three string manipulation functions with syntax and example. | 06 | L2 | 4 | 1 |
| b) | Discuss types of function calls with C program. | 06 | L2 | 3 | 1 |
| c) | Build a C function to perform linear search on a given 1-D array. | 04 | L3 | 3 | 2 |
| 6. a) | Discuss various categories of functions with suitable examples for each. | 08 | L2 | 3 | 1 |
| b) | Outline a C program using functions to read two matrices $A(m \times n)$ and $B(p \times q)$ and to compute the product of A and B if the matrices are compatible for multiplication. | 08 | L3 | 3 | 2 |

Unit – III

- | | | | | | |
|-------|---|----|----|---|---|
| 7. a) | Define structures. How to define a structure in C. Explain two ways of declaring structure variables. | 06 | L2 | 5 | 1 |
| b) | Explain the following file handling functions with syntax and example. | | | | |
| i) | <code>fopen()</code> | | | | |
| ii) | <code>fscanf()</code> | | | | |
| iii) | <code>getc()</code> | 06 | L2 | 5 | 2 |
| c) | Discuss benefits of pointers in C. | 04 | L1 | 5 | 1 |
| 8. a) | Define a pointer in C. How it is declared and initialized?. Outline a C program to read n integers into an array and find the sum of elements using pointers. | 08 | L2 | 5 | 2 |
| b) | Discuss input output operations on a file using a suitable C program. | 08 | L3 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome